

Raw Histories or Compact Profiles? A Cost-Aware Empirical Study of User History Representations for Personalized Review Generation with LLMs

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Abstract—Personalized generation with large language models requires not only high-quality outputs but also cost-effective use of user history. We present a focused empirical study of personalized review generation on Amazon Reviews 2023, comparing three deployment-compatible user-history representations: raw-history few-shot prompting, compact user profiles, and a deployment-oriented global parameter-efficient fine-tuning (PEFT) baseline. We evaluate these methods using automatic metrics, estimated API costs, user-retrieval analysis, and a human preference study. Raw-history few-shot prompting achieves the strongest automatic scores, but its gains saturate as more reviews are added while inference cost increases. Compact profiles preserve much of the semantic and perceived personalization quality while using substantially shorter prompts and lower operational cost. In contrast, a single global PEFT adapter without user-specific inference-time context shows limited personalization ability in this setting. We do not claim to introduce a new personalization architecture or a comprehensive cross-domain benchmark; rather, our results clarify practical cost-quality trade-offs for product-review generation under API-style deployment constraints.

I. INTRODUCTION

Large language models (LLMs) have demonstrated strong performance across a wide range of text generation tasks. Recent work has increasingly studied personalized generation with LLMs, where outputs are tailored to individual users [1], [2]. In personalized review generation, a model must not only produce a plausible review for a target item, but also reflect user-specific characteristics such as vocabulary, preferences, sentiment patterns, and writing style.

A common approach to personalization is to provide a user’s past reviews as in-context examples via few-shot prompting or retrieval-augmented generation [3], [4]. This approach is simple and often effective, but the prompt length, inference cost, and latency increase as more history is included. Another option is to summarize the user history into a compact profile [5], which may preserve useful personalization cues with fewer tokens. Parameter-efficient fine-tuning (PEFT) provides a different way to adapt models, but it remains unclear whether a single global adapter can capture fine-grained user-level traits without user-specific context at inference time [6].

These trade-offs are important in practical LLM applications, where personalization is not only a modeling problem but

also a cost-optimization problem. Modern LLM APIs differ substantially in price across model families, and stronger models can be much more expensive than smaller alternatives [7]. This motivates studying whether effective personalization can be achieved by changing how user history is represented, rather than simply using larger models or longer prompts.

In this work, we ask whether raw user histories are necessary for personalized review generation under a cost constraint. Using Amazon Reviews 2023, we compare three user-history representations: raw-history few-shot prompting, compact user profiles, and a deployment-oriented global PEFT baseline. Our contribution is not a new personalization architecture. Instead, this short paper provides: (i) a cost-aware comparison of common user-history representations under a shared review-generation setup, (ii) an analysis of how performance changes as more user reviews are included, and (iii) a small-scale human preference study comparing perceived user-likeness. We find that raw-history prompting gives the strongest automatic scores, but compact profiles are selected at rates close to raw histories in human evaluation while using much shorter prompts. We therefore frame our conclusions as evidence about product-review generation within Amazon Reviews 2023, rather than as a broad claim about all personalization tasks.

a) Related work. Personalization has recently been studied as a general capability of LLMs, with benchmarks and surveys covering personalized classification, recommendation, and generation [1], [2]. Existing work commonly conditions a frozen or prompted LLM on retrieved user histories, past writings, or natural-language profiles. LaMP evaluates retrieval-based personalization across multiple tasks [2], while Guided Profile Generation and prompt rewriting improve how personal context is summarized and presented to LLMs [4], [5]. Personalized review and explanation generation has also been studied in recommender systems, from attribute-to-sequence review generation and neural tips generation [8], [9] to transformer- and retrieval-based explainable recommendation models [10], [11]. More directly, Review-LLM aggregates user histories and ratings and fine-tunes LLMs for personalized review generation [12]. Our work is complementary to these studies: rather than proposing a new generator or a full horizontal benchmark, we

isolate how raw histories, compact profiles, and global PEFT trade off personalization quality against API cost. Stronger personalization systems such as retrieval-selected histories, hybrid profile-plus-retrieval prompts, Review-LLM-style fine-tuning, PEFT+profile inference, profile-conditioned PEFT, and user-specific adapters occupy important but operationally different points in the design space. We discuss these as future comparisons instead of treating our baseline set as exhaustive. This focus also connects to prompt-compression work, which reduces long-context cost and latency [13]–[15], and to personalized PEFT, which stores user-specific behavior in lightweight modules [6].

II. TASK AND COMPARED METHODS

Given a user u , a target item $x_{u,t}$, and the user’s previous k reviews $H_{u,t}^{(k)}$, the task is to generate a review $\hat{y}_{u,t}$ that matches both the target item and the user’s writing style:

$$\hat{y}_{u,t} = f_{\theta}(S, x_{u,t}, z_{u,t}),$$

where S is the system prompt and $z_{u,t}$ is the user-history representation. We compare three ways of constructing $z_{u,t}$ from the same history.

a) Few-shot prompting.: The raw previous reviews $H_{u,t}^{(k)}$ are directly included in the prompt as in-context examples.

b) Compaction.: The same history is first summarized into a compact profile $c_{u,t}^{(k)} = C(H_{u,t}^{(k)})$, which describes the user’s style, preferences, title style, and typical structure. The generated review is conditioned on this profile instead of the raw reviews.

c) Deployment-oriented global PEFT.: We train a single global adapter on review-generation examples from all users. This baseline is intended to test whether shared task/domain adaptation can substitute for user-specific inference-time context in a multi-user deployment. It is not an upper-bound PEFT personalization method: no separate adapter is trained for each user. At inference time, PEFT-only receives no user history, while PEFT+history additionally receives the same in-context history as few-shot prompting.

III. EXPERIMENTAL SETUP

We evaluate personalized review generation on the Amazon Fashion subset of Amazon Reviews 2023. We select users with at least 50 reviews and split each user’s reviews chronologically into 30 training, 10 validation, and 10 test reviews. User histories are constructed only from the training split, and we verify that no history review occurs after the timestamp of the target test review. The final evaluation set contains 749 reviews. In the few-shot and compaction settings, k past reviews of the same user are used as history ($k \in \{1, 5, 10, 30\}$). For each target instance, we select the latest k reviews in the user’s training split; histories are not randomly sampled or retrieved by target-item similarity. All methods used the same output format and decoding settings: JSON fields for title and text, temperature 1.0, top-p 1.0. We did not set an additional task-specific maximum output length beyond the model/API default.

For PEFT, we train a single global adapter using the same training split and evaluate PEFT-only and PEFT+history. PEFT+history tests whether global adaptation and in-context user history are complementary. Fine-tuning is performed using Microsoft Foundry¹. Because the hosted fine-tuning API does not expose all internal adapter and optimizer details, this setting should be interpreted as an API-level deployment baseline rather than a transparent analysis of PEFT mechanisms.

We evaluate gpt-5.4 as the strongest prompting model available in our evaluation setup and gpt-4.1, gpt-4.1-mini, and gpt-4.1-nano as Azure fine-tuning-compatible models. In few-shot and compaction settings, the same past reviews, item metadata, and output format are used for each k . In the compaction setting, gpt-4.1-mini first summarizes the history into a user profile of about 200 tokens, which is then used by the other models in their prompts. All prompts are provided in Appendix B.

For automatic evaluation, we concatenate the title and text of both the generated and reference reviews into a single string and compare them using BLEU, ROUGE-L, Token F1, and Cosine Similarity. Cosine Similarity is computed between embedding vectors obtained from text-embedding-3-large.

We conduct a human evaluation using the Yahoo! Crowdsourcing² platform. In each question, annotators see the same user’s past reviews and select the generated review that is most representative of the user. We prepare 120 questions, collect three judgments per question with randomized system order, and report question-level bootstrap confidence intervals. The human study compares zero-shot, few-shot, and compaction outputs generated by the same gpt-5.4 model. We omit PEFT from this study because the fine-tuning-compatible models in our setup belonged to a different model family; including them would conflate user-history representation with model-family effects. Because annotators evaluate English reviews through a Japanese crowdsourcing interface, we treat the results as perceived personalization judgments rather than definitive measures of writing style.

We use Amazon Fashion as the main evaluation domain and report an additional automatic evaluation on Beauty and Personal Care in Appendix D to assess whether the main trends hold in another product category.

IV. RESULTS

A. Automatic Evaluation

Table I summarizes representative operating points that illustrate the main cost-quality trade-offs; full within-model results are provided in Appendix D and visualized in Figure 2. We use these within-model comparisons to interpret representation effects, while treating cross-model rows in the table as practical deployment operating points rather than controlled model comparisons. Across the selected settings, gpt-5.4 few-shot prompting with $k = 30$ achieves the best automatic scores, obtaining the highest ROUGE-L, Token F1, cosine similarity,

¹<https://azure.microsoft.com/en-US/products/ai-foundry>

²<https://crowdsourcing.yahoo.co.jp/>

Method	Preprocessing	Inference-time input	Cost counted
Few-shot	Select latest k training reviews from the same user	Item metadata + raw review history	Generation prompt tokens
Compaction	Summarize the same k reviews into a compact profile	Item metadata + compact profile	Summarization + generation; cached-profile variant excludes repeated summarization
Global PEFT-only	Train one shared adapter on all users' training reviews	Item metadata only	Inference + amortized training cost
Global PEFT+history	Same shared adapter as PEFT-only	Item metadata + raw review history	Inference + amortized training cost

Fig. 1. Overview of the compared pipelines. User-specific information enters at inference time for few-shot prompting, compaction, and PEFT+history, but only through global training for PEFT-only.

TABLE I
REPRESENTATIVE AUTOMATIC EVALUATION AND COST RESULTS ON AMAZON FASHION ($N = 749$). COSTS INCLUDE SUMMARIZATION FOR COMPACTION AND AMORTIZED TRAINING COST FOR PEFT. FULL COST DETAILS ARE IN APPENDIX A.

Method	Model	k	ROUGE-L	Token F1	Cosine	BLEU	Avg. operational cost (\$)
Zero-shot	GPT-5.4	0	0.1364	0.2418	0.5553	0.0108	0.002920
Few-shot	GPT-5.4	10	0.1793	0.3215	0.6148	0.0405	0.010194
Few-shot	GPT-5.4	30	0.1915	0.3338	0.6288	0.0487	0.019773
Compaction	GPT-4.1	30	0.1195	0.2241	0.6028	0.0107	0.007022
Compaction	GPT-5.4	30	0.1466	0.2759	0.6134	0.0178	0.008329
PEFT-only	GPT-4.1	0	0.1267	0.2360	0.4996	0.0142	0.134169
PEFT+history	GPT-4.1	30	0.1570	0.2830	0.5342	0.0351	0.160260
PEFT+history	GPT-4.1-mini	30	0.1311	0.2450	0.4996	0.0188	0.032009

and BLEU. However, increasing the history length from $k = 10$ to $k = 30$ yields only modest gains relative to the increase in average request cost, suggesting diminishing returns from simply adding more raw user reviews to the prompt.

Compaction produces lower lexical-overlap scores than few-shot prompting, but it preserves semantic similarity well. In particular, compared with gpt-5.4 few-shot prompting with $k = 30$, gpt-4.1 compaction with $k = 30$ requires about 64% lower average operational cost, although its cosine similarity is lower. This indicates that compact user profiles can serve as a cost-efficient representation of user history in this product-review setting, especially when semantic similarity rather than exact lexical overlap is prioritized.

In contrast, global PEFT without user-specific inference-time context shows limited effectiveness. PEFT-only with gpt-4.1 performs poorly in cosine similarity compared with the history-based prompting methods. Although adding in-context history to the fine-tuned model improves some lexical metrics, it does not close the semantic-similarity gap. These results suggest that global adaptation alone is insufficient for capturing fine-grained user-level traits without user-specific inference-time history.

We additionally evaluated whether generated reviews preserve user-identifiable characteristics through a user-retrieval task. For each generated review, we ranked candidate users by cosine similarity between the generated-review embedding and each user-history centroid embedding using text-embedding-3-large. Few-shot prompting achieved the strongest retrieval performance, while compaction remained substantially above zero-shot but below raw-history prompting. This metric is only a proxy for personalization and may favor

raw-history prompting because the retrieval target is derived from history embeddings. Detailed MRR and Recall@K results are reported in Appendix D.

B. Cost-Performance Trade-off

We evaluate whether performance gains are cost-efficient using a fixed price snapshot at evaluation time under standard API pricing, with no cached-input discount and no Batch/Flex discount. Per-request inference cost is:

$$C_{\text{inf}} = \frac{p_{\text{in}}T_{\text{in}} + p_{\text{out}}T_{\text{out}}}{10^6}.$$

Here, p_{in} and p_{out} denote input and output prices per one million tokens, and T_{in} and T_{out} denote the corresponding token counts.

Operational cost depends on the method. Prompting-based methods use $C_{\text{op}} = C_{\text{inf}}$. Compaction includes both summarization and generation. For PEFT, we report amortized cost:

$$C_{\text{op}} = C_{\text{inf}} + \frac{C_{\text{train}}}{N}, \quad (N = 749).$$

Here, C_{train} is the one-time training charge and N is the number of evaluation requests over which it is amortized.

For a conservative estimate, compact profiles are recomputed for each request, although caching would further reduce cost.

Figure 2 shows that few-shot (raw-history) prompting improves similarity but becomes increasingly expensive as k grows, especially for stronger models requiring long prompts.

Compaction achieves a better trade-off in many cases: despite the extra summarization step, its shorter final prompt reduces

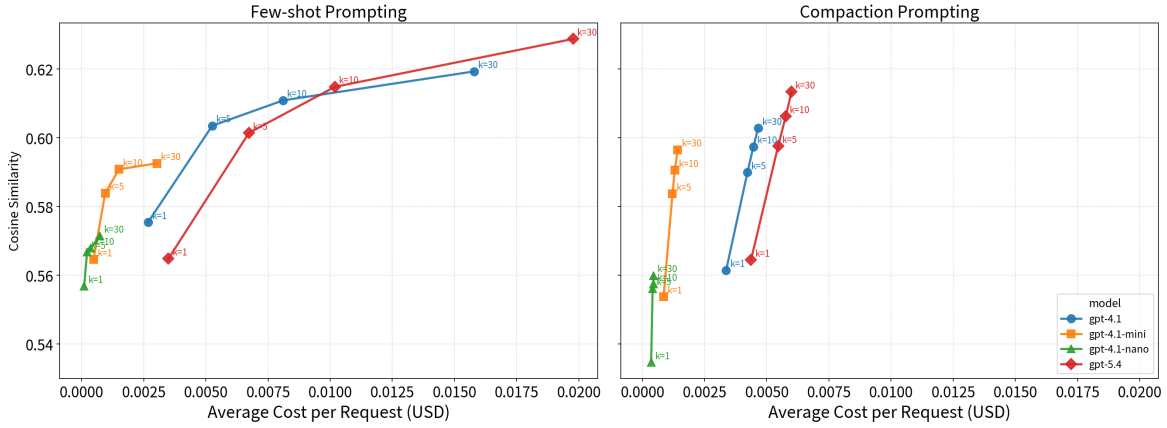


Fig. 2. Cost vs. cosine similarity across different numbers of past reviews k .

overall cost. It reaches comparable cosine similarity to few-shot prompting at substantially lower cost.

For example, compared with gpt-5.4 few-shot prompting with $k = 30$, compaction with gpt-4.1 reduces average operational cost by approximately **64%** under the conservative recomputation assumption. If compact profiles are cached and reused, the $k = 30$ compaction cost drops to the generation-only cost, approximately \$0.0053 for gpt-5.4 and \$0.0040 for gpt-4.1 per request in our token statistics.

C. Human Evaluation

In the human evaluation, we compared zero-shot, few-shot, and compaction outputs generated by gpt-5.4 with $k = 30$, and additionally report question-level bootstrap confidence intervals to account for the grouped annotation structure.

Table II shows the results of the human evaluation on 120 questions. Few-shot prompting received the largest share of votes, followed closely by compaction, while zero-shot was lower.

TABLE II
HUMAN EVALUATION RESULTS ON 120 QUESTIONS. ANNOTATORS
SELECTED THE GENERATED REVIEW THAT BEST REPRESENTED THE USER.

Method	Votes	Share
Few-shot	136	37.8%
Compaction	129	35.8%
Zero-shot	95	26.4%
Total	360	100.0%

A question-level bootstrap analysis showed similar mean vote shares for few-shot (37.8%) and compaction (35.8%), while zero-shot was lower (26.4%). The 95% bootstrap confidence interval for the difference between compaction and few-shot included zero (−11.1 to +6.9 percentage points), indicating that we did not detect a statistically significant difference between the two methods in this sample. In contrast, few-shot showed a significant advantage over zero-shot in an exact binomial test at the question level, while the advantage of compaction over zero-shot was positive but not statistically significant.

The human evaluation results differ from the automatic evaluation. Although few-shot achieved the strongest automatic scores, compaction was preferred at a comparable level by annotators. One possible explanation is that automatic metrics mainly measure similarity to a single reference review, whereas annotators judge whether the output reflects user preferences and writing style more broadly. These results suggest that compact user profiles can preserve much of the perceived personalization quality of raw-history prompting while reducing inference cost in this setting. We interpret this result as indicative rather than conclusive, given the limited scale of the human evaluation and the absence of independent attention checks or language-proficiency screening beyond the crowdsourcing platform’s standard controls.

V. CONCLUSION

We compared three deployment-compatible user-history representations for personalized review generation: raw histories in few-shot prompting, compact user profiles, and a deployment-oriented global PEFT baseline. On Amazon Reviews 2023, few-shot prompting achieved the strongest automatic scores, but its gains saturated as more history was added and required longer and more expensive inputs. Compaction preserved much of the semantic similarity of few-shot prompting at lower inference cost, and human evaluation on 120 questions showed that compaction and few-shot prompting were preferred at comparable rates, with no statistically significant difference detected between them in our question-level analysis. This suggests that compact user profiles can retain much of the perceived personalization quality of raw-history prompting while offering a favorable cost-quality trade-off in product-review generation. Global PEFT showed limited gains in our setting, likely because a single adapter shared across users captures general review patterns more readily than fine-grained user-specific traits.

VI. LIMITATIONS

This study has several limitations. First, both evaluated product categories come from Amazon Reviews 2023. Although

Appendix D includes an additional automatic evaluation on Beauty and Personal Care, the paper is not a broad cross-dataset or cross-task personalization benchmark. Second, the baseline set is intentionally narrow. Retrieval-selected histories, hybrid profile-plus-retrieval methods, Review-LLM-style systems, PEFT+compact profiles, profile-conditioned PEFT, and user-specific adapters are important future comparisons. Third, the PEFT setting uses global adaptation rather than user-specific adapters, which may underestimate the potential of fine-tuning for personalization. The hosted fine-tuning API also limits analysis of internal training mechanisms. Fourth, the model set is limited to the OpenAI/Azure models available in our deployment setup, so the conclusions should not be interpreted as provider-independent evidence across Claude, Gemini, or open-source LLMs. Fifth, BLEU, ROUGE-L, Token F1, embedding cosine similarity, and user-retrieval metrics do not separately measure item faithfulness, rating consistency, aspect preference, or writing-style similarity. Larger human evaluations with targeted criteria would provide stronger evidence. Finally, cost estimates depend on API prices, deployment settings, cached-input discounts, whether compact user profiles are recomputed or reused, and how many downstream requests the one-time PEFT training cost is amortized over.

VII. ETHICAL CONSIDERATIONS

This work studies personalized review generation as a controlled research task and does not advocate deploying systems that generate reviews on behalf of users. Such systems could be misused for deceptive review generation, impersonation, or platform manipulation. Any deployment should require explicit user consent, clear disclosure that the text is AI-generated, and safeguards against automated review spam.

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Item	Description
Dataset	Amazon Reviews 2023
Evaluation domain	Amazon_Fashion
User selection	Users with 50+ reviews per user were selected.
Data split	For each user, reviews were split chronologically into 30 for training, 10 for validation, and 10 for testing.
Evaluation set	749 reviews were sampled from the Amazon_Fashion test split for evaluation.
History definition	The user history $H_{u,t}$ was constructed using only Amazon_Fashion reviews from the training split of the same user.
Future leakage check	No instances were found where a training review in the history occurred after the timestamp of the target test review.
Few-shot history length	$k \in \{1, 5, 10, 30\}$
Few-shot construction	The latest k reviews were selected from the user's Amazon_Fashion training split; no random or target-item retrieval was used.
Item metadata used in prompts	Product title, average rating, product features, product description, product image URLs, store/brand, and product details
Generation target	Review title and review text
Evaluation text	The title and text were concatenated into a single string (title + text) for evaluation.
Output format	Structured JSON format: {"title": ..., "text": ...}

TABLE III
EXPERIMENTAL SETUP.

Past Reviews Summarization Prompt

Instructions

Summarize the following user reviews into a compact user profile for personalized review generation.

Focus on:

- writing style
- common phrases
- preferences
- typical structure

Keep it about 200 tokens.

Input

- User ID: [user_id]
- Past Reviews : [prev_reviews]

Review Generation Prompt - Compaction

Goal

- Generate a new review that reflects the user's style, based on user review style and the product information.

Instructions

- Please generate a new review using the following information.

User Style

- Writing style: [user_writing_style]
- Preferences: [preferences]
- Negative review pattern: [negative_review_pattern]
- Title style: [title_style]
- Typical structure: [typical_structure]

Information about the product to be reviewed

- Product name: [title]
- Average rating: [average_rating]
- Product features: [features]
- Product description: [description]
- Product image URLs: [image_urls]
- Store: [store]
- Product details: [details]

Please generate a new review.

Review Generation Prompt - Few-shot

Goal

- Generate a new review that reflects the user's style, based on user review style and the product information.

Instructions

- Please generate a new review using the following information.

User's Past Reviews

- User ID: [user_id]
- Past reviews: [prev_reviews]

Information about the product to be reviewed

- Product name: [title]
- Average rating: [average_rating]
- Product features: [features]
- Product description: [description]
- Product image URLs: [image_urls]
- Store: [store]
- Product details: [details]

Please generate a new review.

Table IV presents qualitative examples comparing raw user histories, compacted user profiles, and generated reviews from few-shot prompting and compaction. These examples illustrate that compaction can preserve salient stylistic patterns and user preferences while using a much shorter representation than raw review history.

TABLE IV
QUALITATIVE EXAMPLE FOR A CONCISE, PRODUCT-FOCUSED USER. BOTH
GENERATED OUTPUTS USE gpt-5.4 WITH $k = 30$.

Item	Content
User ID	U1
Raw reviews	(1) Misleading description “They need to change the description to item is a SHIRT!” (2) Super cute “Fits perfectly, thick material, vibrant color.” (3) Pretty nightgown “Fits as expected, very pretty and daughter loves wearing it.”
Generated profile	Writing style: concise, plainspoken, and product-focused; mostly positive, with short descriptive sentences. Preferences: good fit, cute appearance, comfort, durability, and vibrant colors. Negative review pattern: directly points out misleading descriptions or construction flaws. Title style: very short titles such as “Super cute.” Typical structure: starts with fit or appearance, then mentions comfort or material quality, and ends with a brief verdict.
Few-shot output	Title: Cute gift set Review: “Super cute gift set for a brother. The tumbler looks nice, the bracelet and keychain are a fun extra, and everything seems made well. Good quality and would make a nice birthday or Christmas gift.”
Compaction output	Title: Cute gift set Review: “Looks nice and exactly as pictured. The tumbler feels sturdy and keeps drinks hot or cold for a good amount of time. The bracelet and keychain are simple but cute, and the whole set makes a great gift for a brother. My brother really liked it and I had no complaints.”
Reference review	Title: Nice gift Review: “Awesome gift for the brother in your life. Items are made well and functional, listed price is fair.”

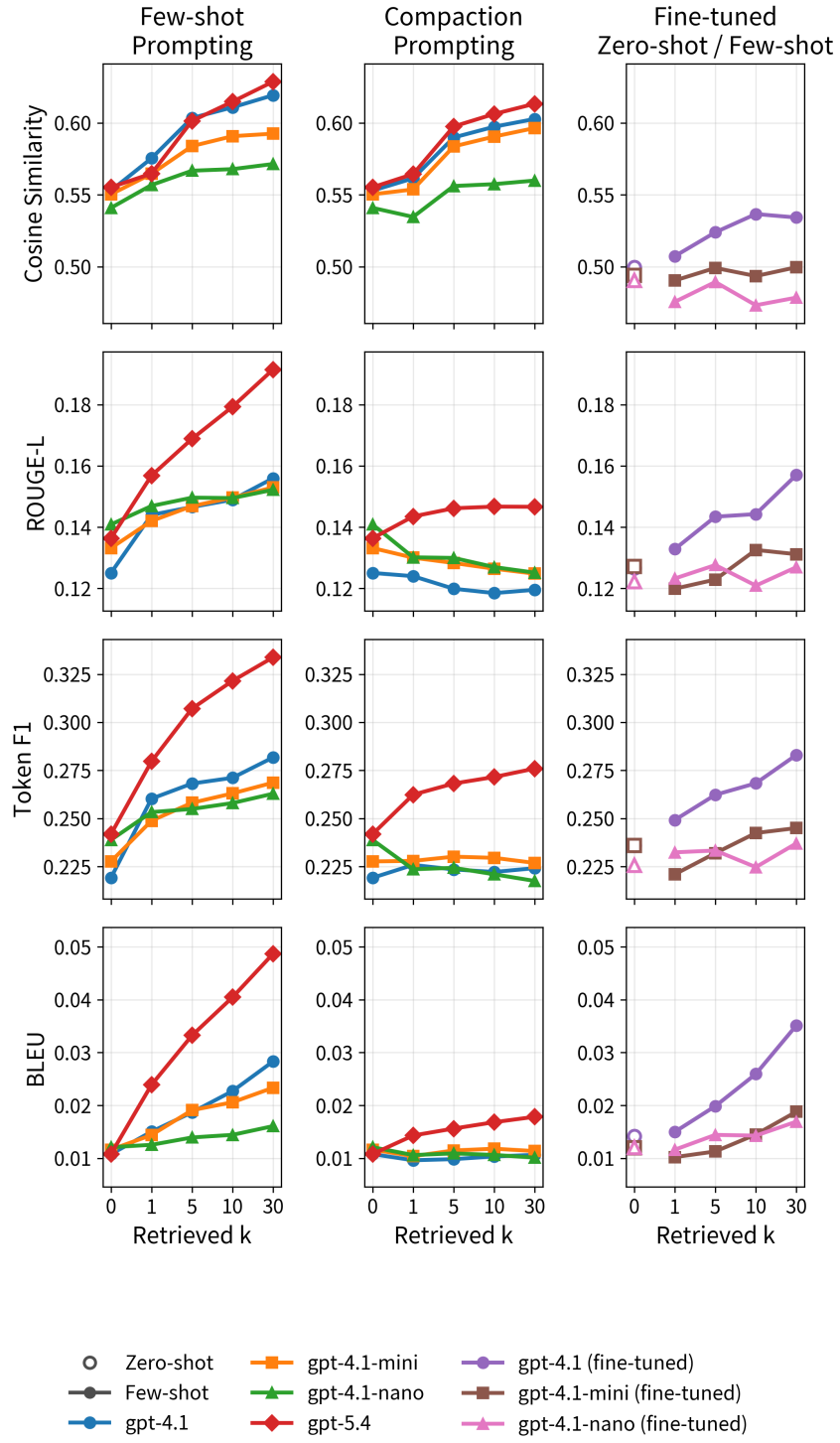


Fig. 3. Automatic evaluation results on the Amazon Fashion domain. For fine-tuned models, Zero-shot denotes PEFT-only inference without user history, while Few-shot denotes PEFT+history inference with in-context past reviews.

TABLE V

USER RETRIEVAL RESULTS ON AMAZON FASHION ($N = 749$ TEST INSTANCES). FOR EACH GENERATED REVIEW, WE RANK CANDIDATE USERS BY COSINE SIMILARITY BETWEEN THE GENERATED-REVIEW EMBEDDING AND EACH USER’S HISTORY CENTROID EMBEDDING. MRR AND RECALL@K ARE AVERAGED OVER THE 749 TEST INSTANCES. HIGHER VALUES INDICATE BETTER PRESERVATION OF USER-SPECIFIC CHARACTERISTICS. BOLD VALUES INDICATE THE BEST SCORE AMONG THE SELECTED SETTINGS.

Method	Model	k	MRR	Recall@1	Recall@5	Recall@10
Zero-shot	GPT-5.4	0	0.0730	0.0134	0.0921	0.1842
Few-shot	GPT-5.4	10	0.3426	0.2296	0.4339	0.5648
Few-shot	GPT-5.4	30	0.3874	0.2777	0.5007	0.6195
Compaction	GPT-4.1	30	0.2262	0.1188	0.3097	0.4459
Compaction	GPT-5.4	30	0.2639	0.1562	0.3458	0.4780

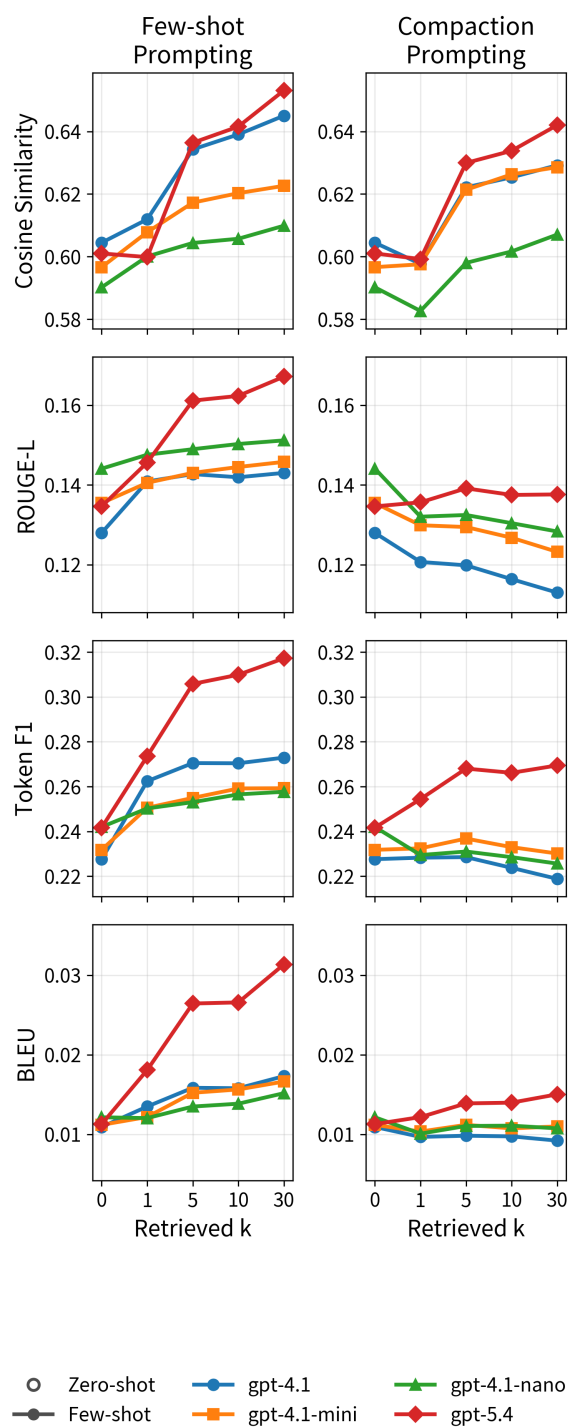


Fig. 4. Automatic evaluation results on the Beauty and Personal Care domain.

We report the exact price snapshot and token statistics used in the Amazon Fashion cost analysis to improve reproducibility. All token counts were recomputed from saved prompts and generations with o200k_base. All costs use standard API pricing with no cached-input discount and no Batch/Flex discount. The per-request inference cost is

$$C_{\text{inf}} = \frac{p_{\text{in}}T_{\text{in}} + p_{\text{out}}T_{\text{out}}}{10^6}.$$

For compaction, we add the summarization-step cost to the final generation cost and conservatively assume that the compact profile is recomputed for every request rather than reused from cache. For PEFT, we report operational cost

$$C_{\text{op}} = C_{\text{inf}} + \frac{C_{\text{train}}}{N}, \quad N = 749.$$

Importantly, the compaction prompt asked the summarization model to keep the profile at about 200 tokens, but this was a soft instruction rather than a hard constraint; therefore, the appendix reports the realized profile lengths observed in the saved prompts.

TABLE VI

PRICE SNAPSHOT USED IN THE COST ANALYSIS. PRICES ARE USD PER 1M TOKENS. THE OPENAI API PRICING PAGE WAS ACCESSED ON 2026-04-26. FINE-TUNING TRAINING COSTS ARE THE ONE-TIME AZURE FOUNDRY CHARGES FOR THE TRAINING RUNS USED IN THIS PAPER, CONVERTED AT 159 JPY/USD.

Role	Model	Input price	Output price	One-time training cost
Generation	gpt-5.4	2.50	15.00	-
Generation	gpt-4.1	2.00	8.00	-
Generation / Summarization	gpt-4.1-mini	0.40	1.60	-
Generation	gpt-4.1-nano	0.10	0.40	-
PEFT inference	gpt-4.1-ft	4.00	16.00	98.06
PEFT inference	gpt-4.1-mini-ft	0.80	3.20	19.61
PEFT inference	gpt-4.1-nano-ft	0.20	0.80	5.88

TABLE VII

AVERAGE TOKEN COUNTS FOR PROMPTING-BASED CONDITIONS ON AMAZON FASHION ($N = 749$). FOR COMPACTION, THE SUMMARIZATION COLUMNS REPORT THE EXTRA USER-PROFILE GENERATION STEP. THE ABOUT-200-TOKEN INSTRUCTION IN THE SUMMARIZATION PROMPT WAS A SOFT INSTRUCTION ONLY AND WAS NOT ENFORCED BY TRUNCATION OR CONSTRAINED DECODING. ALL TOKEN COUNTS WERE COMPUTED WITH o200k_base.

Method	Model	k	Avg. gen input	Avg. gen output	Avg. summary input	Avg. summary output
zero-shot	gpt-5.4	0	305.0	143.8	-	-
zero-shot	gpt-4.1	0	305.0	216.5	-	-
zero-shot	gpt-4.1-mini	0	305.0	162.2	-	-
zero-shot	gpt-4.1-nano	0	305.0	121.2	-	-
few-shot	gpt-5.4	1	621.6	129.3	-	-
few-shot	gpt-5.4	5	1748.6	157.7	-	-
few-shot	gpt-5.4	10	3112.1	160.9	-	-
few-shot	gpt-5.4	30	6918.2	165.2	-	-
few-shot	gpt-4.1	1	621.6	178.9	-	-
few-shot	gpt-4.1	5	1748.6	220.0	-	-
few-shot	gpt-4.1	10	3112.1	235.5	-	-
few-shot	gpt-4.1	30	6918.2	243.5	-	-
few-shot	gpt-4.1-mini	1	621.6	145.4	-	-
few-shot	gpt-4.1-mini	5	1748.6	159.2	-	-
few-shot	gpt-4.1-mini	10	3112.1	163.3	-	-
few-shot	gpt-4.1-mini	30	6918.2	163.8	-	-
few-shot	gpt-4.1-nano	1	621.6	111.4	-	-
few-shot	gpt-4.1-nano	5	1748.6	121.4	-	-
few-shot	gpt-4.1-nano	10	3112.1	121.0	-	-
few-shot	gpt-4.1-nano	30	6918.2	124.4	-	-
compaction	gpt-5.4	1	498.9	174.1	333.9	177.5
compaction	gpt-5.4	5	547.9	229.5	1460.9	226.5
compaction	gpt-5.4	10	563.8	244.4	2824.5	242.3
compaction	gpt-5.4	30	577.2	255.0	6630.5	255.7
compaction	gpt-4.1	1	463.4	253.9	333.9	142.0
compaction	gpt-4.1	5	513.2	331.2	1460.9	191.8
compaction	gpt-4.1	10	529.1	351.2	2824.5	207.7
compaction	gpt-4.1	30	550.9	362.6	6630.5	229.5
compaction	gpt-4.1-mini	1	453.4	181.4	334.1	131.9
compaction	gpt-4.1-mini	5	533.1	244.9	1460.9	211.7
compaction	gpt-4.1-mini	10	553.9	260.8	2824.5	232.6
compaction	gpt-4.1-mini	30	581.3	279.5	6630.5	259.9
compaction	gpt-4.1-nano	1	406.2	155.6	333.9	84.5
compaction	gpt-4.1-nano	5	421.4	193.8	1460.9	100.1
compaction	gpt-4.1-nano	10	425.3	239.8	2824.5	103.9
compaction	gpt-4.1-nano	30	426.6	241.6	6630.5	105.2

TABLE VIII

AVERAGE TOKEN COUNTS FOR PEFT CONDITIONS ON AMAZON FASHION. THE LAST COLUMN REPORTS THE ONE-TIME TRAINING COST AMORTIZED ACROSS $N = 749$ EVALUATION REQUESTS, MATCHING THE OPERATIONAL-COST DEFINITION USED IN TABLE 1.

Method	Model	k	Avg. input	Avg. output	C_{train}/N (USD)
PEFT-only	gpt-4.1 (FT-global)	0	306.1	126.5	0.130919
PEFT-only	gpt-4.1-mini (FT-global)	0	306.1	123.8	0.026184
PEFT-only	gpt-4.1-nano (FT-global)	0	306.1	129.5	0.007855
PEFT+history	gpt-4.1 (FT-global)	1	577.3	123.0	0.130919
PEFT+history	gpt-4.1 (FT-global)	5	1608.2	138.7	0.130919
PEFT+history	gpt-4.1 (FT-global)	10	3006.0	135.1	0.130919
PEFT+history	gpt-4.1 (FT-global)	30	6772.8	140.6	0.130919
PEFT+history	gpt-4.1-mini (FT-global)	1	577.3	130.3	0.026184
PEFT+history	gpt-4.1-mini (FT-global)	5	1608.2	130.2	0.026184
PEFT+history	gpt-4.1-mini (FT-global)	10	3006.0	115.0	0.026184
PEFT+history	gpt-4.1-mini (FT-global)	30	6772.8	127.3	0.026184
PEFT+history	gpt-4.1-nano (FT-global)	1	577.3	131.4	0.007855
PEFT+history	gpt-4.1-nano (FT-global)	5	1608.2	154.3	0.007855
PEFT+history	gpt-4.1-nano (FT-global)	10	3006.0	152.0	0.007855
PEFT+history	gpt-4.1-nano (FT-global)	30	6772.8	144.8	0.007855

TABLE IX

PEFT OPERATIONAL-COST SENSITIVITY TO THE NUMBER OF AMORTIZED REQUESTS N . EACH CELL REPORTS $C_{op}(N) = C_{inf} + C_{train}/N$ UNDER THE SAME STANDARD PRICING ASSUMPTION USED IN TABLE 1, WITH NO CACHED-INPUT DISCOUNT AND NO BATCH/FLEX DISCOUNT.

Method	Model	k	$N = 749$	$N = 5,000$	$N = 10,000$	$N = 50,000$
PEFT-only	gpt-4.1	0	0.134169	0.022861	0.013055	0.005210
PEFT+history	gpt-4.1	30	0.160260	0.048952	0.039146	0.031302
PEFT+history	gpt-4.1-mini	30	0.032009	0.009748	0.007787	0.006218

TABLE X

CACHED-PROFILE SENSITIVITY FOR COMPACTION AT $k = 30$ ON AMAZON FASHION. CONSERVATIVE COST RECOMPUTES THE PROFILE FOR EVERY REQUEST; CACHED COST COUNTS ONLY FINAL REVIEW GENERATION AFTER A PROFILE HAS ALREADY BEEN PRODUCED.

Model	Conservative cost	Cached-profile cost
gpt-5.4	0.008329	0.005268
gpt-4.1	0.007022	0.004003

TABLE XI

AVERAGE RAW-HISTORY LENGTH AND COMPACT-PROFILE LENGTH ON AMAZON FASHION. RAW-HISTORY LENGTH MEASURES ONLY THE HISTORY BLOCK IN THE FEW-SHOT PROMPT; FULL FEW-SHOT PROMPT LENGTH IS SHOWN SEPARATELY BECAUSE ITEM METADATA IS SHARED ACROSS CONDITIONS. THE PROFILE LENGTH STATISTICS REFLECT THE REALIZED PROFILES ACTUALLY INSERTED INTO THE GENERATION PROMPT, NOT A HARD 200-TOKEN CAP.

k	Raw history mean	Raw history std.	Full few-shot prompt mean	Profile mean	Profile std.
1	254.7	174.3	621.6	134.0	42.2
5	1381.8	753.7	1748.6	182.5	57.6
10	2745.3	1407.9	3112.1	196.6	63.9
30	6551.3	3728.7	6918.2	212.6	72.4

The human evaluation used 120 questions, three independent judgments per question, and randomized system order. We report question-level bootstrap confidence intervals in the main text to avoid treating the three judgments for the same question as fully independent. The task was distributed through Yahoo! Crowdsourcing’s standard worker pool and interface. We did not independently collect annotator English proficiency, add separate attention-check questions, or compute a reliable agreement statistic from additional gold labels. These omissions limit the strength of the human-evaluation conclusions, so we interpret the study as a small perceived-personalization check rather than a definitive assessment of user style.

【あるユーザーの過去レビュー】

「These tights fit my 4yo great! She has room to grow with these tights. These tights are meant for a girl much larger than a 4yo. I would suggested ordering up at least one size. As you can see from the pictures these tights are quite see through. They also have a high rise since they are meant for an older child. These tights are comfortable and hold up well to the abuse of a 4yo girl.」

「These sunglasses don't feel like they would do much for protection from the sun. They are oversized as stated. They are NOT mirrored as stated. As you can tell - you can see right through them. They feel very 80s to me. I wouldn't wear these for anything but with a costume. They feel cheap like something from a gas station. One of the sides doesn't stay open and is floppy. Maybe I'll have an 80s theme party I can use these for!」

「This jacket is very nice looking and fits my 7-year-old nephew really good. He was really interested in the compass on the sleeve but it was hard for him to use since it was so high up on the sleeve, it would have been better placed by his wrist. The care instructions are very hard to read, but it says not to put into the dryer and wash cold. I was disappointed that the chest zipper doesn't go to a real pocket. The lining is very soft. The hood is detachable via zipper. There is velcro on the hood to keep it on the head, but instead of connecting to each other from each side, the velcro attaches to the main body of the coat, which was interes」

【レビューA】

タイトル : Cute Statement Earrings With a Trendy Look

レビュー : These earrings have a fun, fashionable design and the mottled light leopard pattern gives them a stylish statement look without being too over the top. The acrylic resin material makes them appear lightweight, which is nice for longer wear. They seem like a good accessory for casual outfits or dressing up a simple look. That said, with the overall average rating being 3.7, they may not be perfect for everyone, but if you like bold hoop-style fashion jewelry, these could be a cute addition to your collection.

【レビューB】

タイトル : Cute look, but a little cheap

レビュー : These are cute statement earrings and the light leopard pattern is fun, but they do look a little cheaper in person than I expected. The size is noticeable without being too over the top, and they're lightweight enough to wear comfortably for a few hours. The acrylic resin has a decent mottled look, but the finish and overall construction aren't amazing. Mine were fine out of the package, but they don't feel especially sturdy, so I wouldn't expect them to hold up to rough handling. The posts and backing were usable, just not great quality. If you want an inexpensive pair for occasional wear, they do the job and add a nice pop to a simple outfit. I took off stars because the material looks a little cheap and the durability doesn't seem like it will be the best.

【レビューC】

タイトル : Cute but look a little cheap up close.

レビュー : These earrings are lightweight and comfortable to wear. The mottled light leopard pattern is cute and fun for summer or fall. They are a good medium statement size without being too heavy on the ears. From far away they look really nice, but up close the acrylic does look a little cheap. The pattern on each earring is a little different which is expected with this type of material. If you're wanting an inexpensive trendy pair of earrings, these are cute. If you want something that looks higher end, these probably aren't it.

【質問】 どのAIのレビューがユーザーのレビュースタイルをもっとも反映していますか？（文字数、表現、文体等）

☐ レビューA

☐ レビューB

☐ レビューC

Fig. 5. Screenshot of the Yahoo! Crowdsourcing interface used for human evaluation. Annotators were shown a user’s past reviews and multiple generated candidate reviews, and selected the output that was most representative of the user’s writing style and preferences.